Notes: Hansen & Tong (RFS, 2026)

Option Pricing with Time-Varying Volatility Risk Aversion

Contents

1 What the paper is doing

The paper builds an option-pricing model in which *volatility risk aversion varies over time* and is inferred directly from derivative prices (VIX and option-implied volatilities), while the physical return dynamics remain in the familiar Heston–Nandi GARCH class. The key object is the *variance risk ratio* (VRR), denoted η_t, defined as the ratio between the one-step-ahead risk-neutral and physical conditional variances. By allowing η_t to move stochastically (rather than being a constant), the model can generate empirically realistic, time-varying shapes of the pricing kernel and substantially improves joint fit to VIX and option panels. Technically, the authors (i) derive the no-arbitrage mapping from a time-varying pricing kernel to risk-neutral dynamics, (ii) obtain a closed-form VIX formula, and (iii) develop an accurate analytic approximation for option prices despite the non-affine state dynamics induced by time-varying risk aversion.

2 Core objects

2.1 Physical vs risk-neutral conditional variance and the VRR

Let R_{t+1} denote the (log) market return from t to $t+1$, and let h_{t+1} be the physical conditional variance:

$$h_{t+1} \equiv \mathrm{Var}_t^{\mathbb{P}}(R_{t+1}).$$

Under the risk-neutral measure $\mathbb{Q}$, the (one-step) conditional variance is

$$h^*_{t+1} \equiv \mathrm{Var}_t^{\mathbb{Q}}(R_{t+1}).$$

The paper defines the **variance risk ratio** (VRR) as

$$\eta_t \;\equiv\; \frac{h^*_{t+1}}{h_{t+1}}. \tag{1}$$

A closely related object is the one-step **variance risk premium** (VRP):

$$VRP_t \;\equiv\; h_{t+1} - h^*_{t+1} \;=\; (1-\eta_t)\,h_{t+1}. \tag{2}$$

Hence, when $\eta_t > 1$ (a common empirical pattern), the risk-neutral variance exceeds the physical variance and $VRP_t < 0$.

2.2 Why letting η_t vary matters

If η_t is constant, then h^*_{t+1} is proportional to h_{t+1} and the risk-neutral variance inherits all of its dynamics from lagged physical returns. This is restrictive because derivative prices (VIX, implied volatility surface) contain information about risk-neutral dynamics that is not spanned by the physical GARCH state alone. Allowing η_t to be time-varying relaxes this proportionality and opens a channel through which derivative prices identify *time variation in volatility risk compensation.*

3 Model 1: Physical return dynamics (Heston–Nandi GARCH under $\mathbb{P}$)

The physical dynamics are the Heston–Nandi GARCH (HNG) model:

$$R_{t+1} = r + \Big(\lambda - \tfrac{1}{2}\Big)h_{t+1} + \sqrt{h_{t+1}}\,z_{t+1}, \qquad z_{t+1} \sim \text{iid } N(0,1), \tag{3}$$

$$h_{t+1} = \omega + \beta h_t + \alpha\big(z_t - \gamma\sqrt{h_t}\big)^2. \tag{4}$$

- r is the one-period risk-free rate (taken as known/observed).
- λ loads the conditional variance into expected returns (a risk-premium parameter).
- $(\omega, \beta, \alpha, \gamma)$ govern the GARCH variance recursion; γ introduces asymmetry ("leverage").

Stationarity/positivity intuition. The conditional variance is positive by construction under standard parameter restrictions and the nonnegativity constraint. A typical persistence measure under $\mathbb{P}$ is

$$\pi^{\mathbb{P}} \approx \beta + \alpha(1 + \gamma^2),$$

which is reported in the empirical table as a convenient summary of volatility persistence.

4 Model 2: Pricing kernel with constant volatility risk aversion (CHNG)

4.1 Pricing kernel (constant parameters)

The Christoffersen–Heston–Jacobs (2013)-style pricing kernel used as the benchmark in the paper is

$$\frac{M_{t+1}}{M_t} = \left(\frac{S_{t+1}}{S_t}\right)^{\phi} \exp\Big\{\delta + \pi h_{t+1} + \xi\big(h_{t+2} - h_{t+1}\big)\Big\}, \tag{5}$$

where (ϕ, δ, π, ξ) are constants. Conditioning on the return filtration $\mathcal{F}_t^R$, the paper works with the normalized kernel

$$M_{t+1,t} \;\equiv\; \frac{M_{t+1}}{\mathbb{E}_t^{\mathbb{P}}[M_{t+1} \mid \mathcal{F}_t^R]} \;=\; \frac{\exp(\phi R_{t+1} + \xi h_{t+2})}{\mathbb{E}_t^{\mathbb{P}}[\exp(\phi R_{t+1} + \xi h_{t+2}) \mid \mathcal{F}_t^R]}. \tag{6}$$

4.2 Risk-neutral dynamics induced by the kernel

Under this constant-kernel specification, the risk-neutral return and variance remain HNG, with parameter mapping:

$$R_{t+1} = r - \tfrac{1}{2}h^*_{t+1} + \sqrt{h^*_{t+1}}\, z^*_{t+1}, \qquad z^*_{t+1} \sim \text{iid } N(0,1), \tag{7}$$

$$h^*_{t+1} = \omega^* + \beta h^*_t + \alpha^*(z^*_t - \gamma^*\sqrt{h^*_t})^2, \tag{8}$$

where

$$h^*_t = \eta\, h_t, \qquad \omega^* = \omega\eta, \qquad \alpha^* = \alpha\eta^2, \qquad \gamma^* = \frac{1}{\eta}\Big(\gamma + \lambda - \tfrac{1}{2}\Big) + \tfrac{1}{2}, \qquad \eta = (1 - 2\alpha\xi)^{-1}. \tag{9}$$

Key restriction: the VRR is constant, $\eta_t \equiv \eta$, so h^*_{t+1} is proportional to h_{t+1} at all dates.

5 Model 3: Dynamic volatility risk aversion (DHNG)

5.1 Dynamic pricing kernel

The paper generalizes the normalized pricing kernel by letting the key preference parameters be time-varying:

$$M_{t+1,t} = \frac{\exp\big(\phi_t R_{t+1} + \xi_t h_{t+2}\big)}{\mathbb{E}_t^{\mathbb{P}}\big[\exp\big(\phi_t R_{t+1} + \xi_t h_{t+2}\big) \mid \mathcal{F}_t^R\big]}, \tag{10}$$

where ϕ_t and ξ_t are $\mathcal{G}_t$-measurable (the broader information set that can include derivative prices).

5.2 Theorem 1: No-arbitrage mapping and the VRR identity

Define

$$\eta_t \equiv (1 - 2\alpha\xi_t)^{-1}. \tag{11}$$

Theorem 1 (core mapping). Under the physical HNG dynamics and the dynamic kernel above, the time-varying parameters satisfy

$$\xi_t = \frac{1}{2\alpha}\frac{\eta_t - 1}{\eta_t}, \tag{12}$$

$$\phi_t = \frac{\eta_t - 1}{\eta_t}\Big(\gamma - \tfrac{1}{2}\Big) - \frac{1}{\eta_t}\lambda, \tag{13}$$

and the risk-neutral dynamics remain HNG *but with time-varying parameters*:

$$R_{t+1} = r - \tfrac{1}{2}h^*_{t+1} + \sqrt{h^*_{t+1}}\, z^*_{t+1}, \tag{14}$$

$$h^*_{t+1} = \omega^*_t + \beta^*_t h^*_t + \alpha^*_t (z^*_t - \gamma^*_t \sqrt{h^*_t})^2, \tag{15}$$

where

$$\omega^*_t = \omega\eta_t, \qquad \beta^*_t = \beta\frac{\eta_t}{\eta_{t-1}}, \qquad \alpha^*_t = \alpha\eta_t\eta_{t-1}, \qquad \gamma^*_t = \frac{1}{\eta_t}\Big(\gamma + \lambda - \tfrac{1}{2}\Big) + \tfrac{1}{2}. \tag{16}$$

Finally,

$$\eta_t = \frac{h^*_{t+1}}{h_{t+1}}, \tag{17}$$

so η_t is exactly the variance risk ratio (not merely a label).

5.3 Lemma 1: Shape of the pricing kernel and why it can flip

A key diagnostic is the log pricing kernel as a function of the return innovation. **Lemma 1** shows that

$$\log M_{t+1,t} = -\lambda R_{t+1} + \xi_t \alpha \frac{(R_{t+1} - r)^2}{h_{t+1}} + \kappa_{0,t} + \kappa_{1,t} h_{t+1}, \tag{18}$$

where

$$\kappa_{0,t} = -\tfrac{1}{2}\log\eta_t, \qquad \kappa_{1,t} = \tfrac{1}{2}\Big(\lambda - \tfrac{1}{2}\Big)^2 - \tfrac{1}{8}\eta_t.$$

Interpretation (the central economic intuition).

- The coefficient on the quadratic term in $(R_{t+1} - r)$ is proportional to ξ_t.
- If $\xi_t > 0$, the pricing kernel is *U-shaped* in returns; if $\xi_t < 0$, it is *inverted U-shaped*.
- Because ξ_t (equivalently η_t) is allowed to move over time, the model can match the empirically observed instability in the pricing-kernel shape (Figure 1).

6 Derivative pricing with dynamic volatility risk aversion

6.1 VIX pricing

Let M denote the number of trading days in the VIX horizon (e.g., $M \approx 22$ for one month). Given the risk-neutral variance sequence, the model implies the M-period-ahead VIX formula:

$$VIX_t = A \times \sqrt{\frac{1}{M}\sum_{k=1}^{M} \mathbb{E}^{\mathbb{Q}}_t\big(h^*_{t+k}\big)}, \qquad A = 100\sqrt{252}. \tag{19}$$

Assumption 3 (ARMA for log VRR). The paper assumes

$$\varphi(L)\big(\log \eta_t - \zeta\big) = \theta(L)\varepsilon_t, \tag{20}$$

where $\varphi(\cdot)$ and $\theta(\cdot)$ are lag polynomials and ε_t is an innovation with mean zero and variance σ_ε^2.

Theorem 2 (closed-form VIX). Under the physical model, the dynamic kernel, and the ARMA specification for $\log \eta_t$, VIX is available in closed form:

$$VIX_t = \sqrt{a_1(M, \sigma_\varepsilon^2) + a_2(M, \sigma_\varepsilon^2)\, h_{t+1}^*}, \tag{21}$$

where (a_1, a_2) depend on the lagged VRR history through the ARMA state and on the moment generating function of ε_t.

Why this is useful: it delivers a tractable VIX measurement equation even though the option model is no longer affine in the full state vector once η_t is stochastic.

6.2 Option pricing

6.2.1 A conditional MGF is the key input

For horizon M define the cumulative return

$$R_{t,M} \equiv \sum_{i=1}^{M} R_{t+i}.$$

Let the conditional MGF under $\mathbb{Q}$ be

$$g_{t,M}(s) \equiv \mathbb{E}_t^{\mathbb{Q}}\big[\exp\big(sR_{t,M}\big)\big]\,. \tag{22}$$

If $g_{t,M}$ is known for complex arguments, European options can be priced via Fourier inversion.

6.2.2 Lemma 2: MGF becomes affine if the future VRR path is treated as given

Fix a *predetermined* future VRR path $\bar{\eta}_{t,M} = (\log \eta_{t+1}, \dots, \log \eta_{t+M})'$ that is measurable at time t. Then **Lemma 2** states:

$$g_{t,M}\big(s \mid \bar{\eta}_{t,M}\big) = \exp\big(A_T^*(s, M) + B_T^*(s, M) h_{t+1}^*\big), \qquad T = t + M, \tag{23}$$

where (A_T^*, B_T^*) solve backward recursions analogous to the Heston–Nandi case.

Economic meaning: conditional on a path for risk aversion, the model is essentially "affine again," so the intractability comes only from uncertainty about the future path of η.

6.2.3 Theorem 3: a second-order analytic approximation for stochastic VRR

Because $\eta_{t,M}$ is random, the paper approximates the MGF by a second-order Taylor expansion around the *risk-neutral conditional mean path* $\bar{\eta}_{t,M}^e \equiv \mathbb{E}_t^{\mathbb{Q}}(\eta_{t,M})$:

$$\widehat{g}_{t,M}(s) = g_{t,M}\big(s \mid \bar{\eta}_{t,M}^e\big) \left(1 + \frac{1}{2}\mathrm{Tr}(H_{t,M}(s)\, \Sigma_M)\right), \tag{24}$$

where $H_{t,M}(s)$ is the Hessian of $g_{t,M}$ with respect to the VRR path and Σ_M is the conditional covariance matrix of the ARMA innovations driving $\log \eta$.

Fourier-based call price (using $\widehat{g}$). Let K be the strike and S_t the spot index level. The approximate call price is

$$\widehat{C}_{t,M} = S_t \,\widehat{P}_{1,t} - K e^{-rM} \widehat{P}_{2,t}, \tag{25}$$

with

$$\widehat{P}_{1,t} = \frac{1}{2} + \frac{e^{-rM}}{\pi} \int_0^\infty \Re\left(\frac{K^{-iu}\,\widehat{g}_{t,M}(iu+1)}{iu\,S_t} \right) du, \tag{26}$$

$$\widehat{P}_{2,t} = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re\left(\frac{K^{-iu}\,\widehat{g}_{t,M}(iu)}{iu} \right) du. \tag{27}$$

6.2.4 Theorem 4: a general moment formula from an MGF

The paper provides a convenient way to compute conditional moments under $\mathbb{Q}$: for $k \in \mathbb{N}$,

$$\mathbb{E}_t^{\mathbb{Q}}\big(R_{t,M}^k\big) = \frac{1}{\pi} \int_0^\infty \Re\left[\frac{k!}{v^{k+1}} g_{t,M}(v) - \frac{k!}{u^{k+1}} g_{t,M}(u) \right] ds, \tag{28}$$

where $u = u_R + is$ and $v = v_R + is$ are complex numbers with $u_R < 0$ and $v_R > 0$.

Intuition: the formula recovers moments by integrating the MGF along vertical lines in the complex plane (a transform-based identity), avoiding repeated numerical differentiation of g at zero.

7 Observation-driven dynamics for volatility risk aversion (estimation model)

7.1 Why an observation-driven (score) update?

Derivative prices are observed at high frequency and directly reflect the market's risk-neutral beliefs and risk compensation. The paper updates $\log \eta_t$ using the *score* of the derivative-price likelihood, so the state moves in the direction that reduces derivative pricing errors. This is conceptually parallel to GARCH volatility updates, but applied to the *volatility risk aversion* state.

7.2 Likelihood decomposition

Let X_t denote the vector of observed derivative objects (either $\log(VIX_t)$ or a panel of log implied vols). The full log-likelihood factors as

$$\ell(R_{1:T}, X_{1:T} \mid \mathcal{F}_0) = \sum_{t=1}^{T} \ell(R_t \mid \mathcal{F}_{t-1}) + \sum_{t=1}^{T} \ell(X_t \mid \mathcal{G}_t), \tag{29}$$

where $\ell(R_t \mid \cdot)$ comes from the Gaussian HNG return shock and $\ell(X_t \mid \cdot)$ is a measurement equation for derivative prices.

7.3 Derivative measurement equation and equicorrelated errors

Let X_t^m be the model-implied derivative vector. The pricing error is $e_t \equiv X_t - X_t^m$. The paper assumes

$$e_t \mid \mathcal{G}_t \sim N\big(0, \sigma_e^2 \Omega_{N_t}\big), \qquad \Omega_{N_t} = (1-\rho) I_{N_t} + \rho\, \mathbf{U}_{N_t}, \tag{30}$$

where N_t is the number of derivative observations at date t, and $\mathbf{U}_{N_t}$ is the all-ones matrix.

Intuition: an equicorrelation structure captures a strong common component in option pricing errors while remaining feasible in an unbalanced panel.

7.4 Score-driven innovations

The ARMA innovation is modeled as

$$\varepsilon_{t+1} = \sigma s_t, \qquad s_t \equiv \frac{\nabla_t}{\sqrt{\mathbb{E}_t^{\mathbb{P}}(\nabla_t^2)}}, \qquad \nabla_t \equiv \frac{\partial \ell(X_t \mid \mathcal{G}_t)}{\partial \log \eta_t}. \tag{31}$$

Theorem 5 (explicit score). Under the Gaussian measurement equation,

$$\nabla_t = \frac{1}{\sigma_e^2} \left(\frac{\partial X_t^m}{\partial \log \eta_t} \right)' \Omega_{N_t}^{-1} e_t, \qquad \mathbb{E}_t^{\mathbb{P}}(\nabla_t) = 0. \tag{32}$$

If X_t^m is scalar, then s_t is essentially a signed, standardized pricing error.

7.5 A concrete DHNG state recursion used in the empirical section

While the theory allows general ARMA(p,q), the empirical implementation uses an AR(1) specification for $\log \eta_t$ (as reflected by Table 3):

$$\log \eta_{t+1} - \zeta = \phi(\log \eta_t - \zeta) + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} = \sigma s_t. \tag{33}$$

Because s_t depends only on information through date t, the state η_{t+1} is predetermined when pricing derivatives at $t+1$.

8 Tables and figures

8.1 Figure 1: pricing kernel shape instability motivates time-varying risk aversion

Figure 1 plots the log ratio of risk-neutral to physical densities against standardized monthly returns. Empirically inferred log ratios (Panel A) sometimes look U-shaped (e.g., 2001–2002) and sometimes inverted-U-shaped (e.g., 2005–2006). The CHNG model (Panel B) enforces a constant quadratic coefficient and thus delivers an always-U-shaped pattern. **Takeaway:** a constant volatility risk aversion parameter ξ cannot reproduce the observed time variation in the shape of the pricing kernel, motivating a time-varying ξ_t (equivalently η_t).

8.2 Figure 2: what changing η does to risk-neutral distributions

Figure 2 compares properties of cumulative returns under two VRR levels (low $\eta = 0.70$ vs high $\eta = 1.70$):

- The log density ratio $\log(f^{\mathbb{Q}}/f^{\mathbb{P}})$ rotates with η, meaning that risk-neutral tail tilts are directly governed by volatility risk aversion.
- The risk-neutral news-impact curve changes: higher η changes how shocks map into future risk-neutral variance.

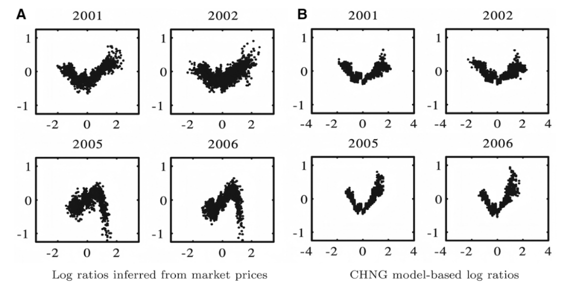

Figure 1
The log ratios of risk-neutral 1-month densities and physical 1-month histogram against log returns in monthly standard deviations. Panels A and B are subsets of figures 3 and 5, respectively, in Christoffersen, Heston, and Jacobs (2013). The former is based on empirical market prices, and the latter has the corresponding CHNG model-based log ratios.

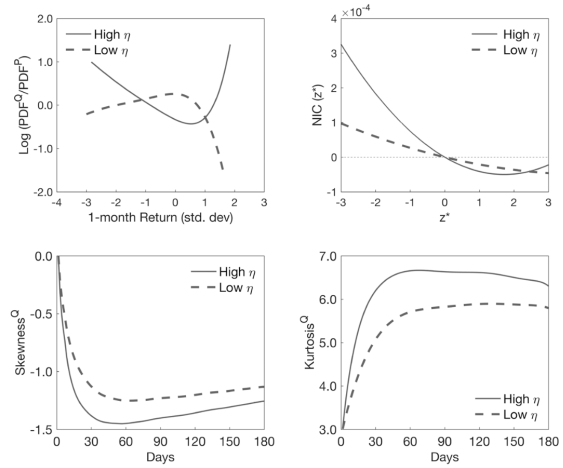

Figure 2
Properties of cumulative returns for two levels of variance risk ratio, $\eta_{\text{low}}=0.70$ (dashed lines) and $\eta_{\text{high}}=1.70$ (solid lines). The upper-left panel presents the log-ratios of the $\mathbb{Q}$-to-$\mathbb{P}$ densities for cumulative returns over 1 month, the corresponding news impact curves under $\mathbb{Q}$ are shown in the upper-right panel, and skewness and kurtosis of multiperiod cumulative returns under the risk-neutral measure $\mathbb{Q}$ are shown in the two lower panels. These results are based on 1 million simulations using a design based on our model estimates (last column in Table 3), where $\eta_{\text{low}}=0.70$ and $\eta_{\text{high}}=1.70$ correspond to the 10% and 90% quantiles of the unconditional distribution of η_t, respectively.

- Multi-horizon skewness is negative and becomes more negative at short horizons; kurtosis rises quickly with horizon and is larger under high η.

Interpretation: η_t is not a cosmetic parameter; it reshapes the entire risk-neutral distribution and thus the implied volatility surface.

8.3 Figures 3–4 and Table 1: accuracy of the analytic approximation

These objects validate the analytic pricing approximation:

- **Figure 3** shows that the second-order DHNG approximation matches the "true" simulated risk-neutral density of 6-month returns very closely for low/mean/high initial η_0.
- **Figure 4** shows the same for variance, skewness, and kurtosis across horizons.
- **Table 1** summarizes implied-volatility approximation errors $e_i = 100(\log IV^{model} - \log IV^{true})$ over a grid of Black–Scholes deltas and maturities. DHNG errors are near zero, while CHNG exhibits large systematic bias (underpricing when η_0 is low and overpricing when η_0 is high), and the first-order "Auxiliary" structure shows a persistent downward bias that grows with maturity.

Why this matters for estimation: the approximation is accurate enough to embed into likelihood-based estimation without relying on heavy simulation inside the optimizer.

8.4 Table 2: data moments (returns, VIX, and option panels)

Table 2 documents three empirical facts that the model must jointly accommodate:

1. Equity returns have substantial negative skewness and very high kurtosis (fat tails).
2. VIX is highly right-skewed with large excess kurtosis (spikes in crises).

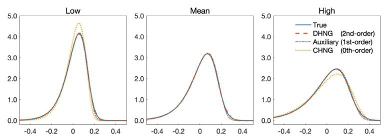

Figure 3
Risk-neutral densities of 6-month cumulative return, for different initial values of variance risk ratio, $\eta_0=0.70$ (left panel), $\eta_0=1.11$ (middle panel), and $\eta_0=1.70$ (right panel). The blue solid line represents the true density from 1 million simulations; the red dashed line is the (2nd-order) approximate density by DHNG; the gray dash-dotted line is the auxiliary model based on $\bar{\eta}^e_{t,M}$ (1st-order); and the yellow dotted line is the density by CHNG with constant η_t (0th-order). The model parameters are based on the estimates with option prices (the last column in Table 3).

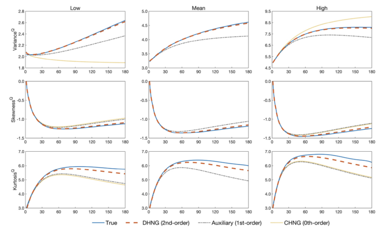

Figure 4
Annualized variance, $(252/M)\mathrm{var}^{\mathbb{Q}}_t(\sum_{i=1}^{M} R_{t+i})\times 100\%$, skewness, and kurtosis of cumulative returns over M days, are shown with three different initial variance risk ratios, $\eta_0=0.70$ (left panels), $\eta_0=1.11$ (middle panels), and $\eta_0=1.70$ (right panels). Blue lines represent the true population moments, red dashed lines are those of DHNG, yellow dotted lines are those for CHNG, and the gray dash-dotted lines are those for the auxiliary "model." Population moments were obtained from 1 million simulated trajectories, using a design based on the parameter estimates obtained with option prices (the last column in Table 3).

Table 1
Approximation errors by logarithm of implied volatility (in percentage)

		Low η			Mean η		High η		
Delta	DTM	CHNG	AUX	DHNG	AUX	DHNG	CHNG	AUX	DHNG
	30	−2.09	−0.41	0.10	−0.48	0.11	1.16	−0.72	−0.02
	60	−5.06	−1.30	0.02	−1.55	−0.04	2.55	−1.69	−0.05
0.3	90	−8.26	−2.23	−0.06	−2.41	−0.07	3.93	−2.56	−0.06
	120	−11.21	−2.99	−0.07	−3.15	−0.06	5.26	−3.28	−0.06
	150	−13.99	−3.70	−0.09	−3.84	−0.09	6.50	−3.93	−0.09
	30	−1.62	−0.02	0.41	−0.49	0.04	1.12	−0.68	−0.07
	60	−4.88	−1.23	−0.04	−1.34	−0.05	2.61	−1.44	−0.05
0.4	90	−7.83	−1.96	−0.07	−2.08	−0.06	4.03	−2.18	−0.06
	120	−10.64	−2.62	−0.05	−2.72	−0.05	5.41	−2.80	−0.05
	150	−13.34	−3.26	−0.07	−3.33	−0.07	6.69	−3.37	−0.07
	30	−1.74	−0.20	0.22	−0.69	−0.18	1.11	−0.63	−0.09
	60	−4.83	−1.21	−0.10	−1.23	−0.05	2.56	−1.32	−0.05
0.5	90	−7.59	−1.82	−0.06	−1.92	−0.06	3.98	−1.99	−0.06
	120	−10.38	−2.46	−0.04	−2.53	−0.04	5.37	−2.57	−0.04
	150	−13.07	−3.07	−0.06	−3.10	−0.06	6.66	−3.11	−0.05
	30	−2.32	−0.76	−0.30	−0.65	−0.17	1.13	−0.55	−0.03
	60	−4.72	−1.11	−0.05	−1.22	−0.05	2.46	−1.29	−0.06
0.6	90	−7.57	−1.82	−0.04	−1.89	−0.05	3.85	−1.94	−0.05
	120	−10.40	−2.46	−0.04	−2.50	−0.03	5.22	−2.51	−0.03
	150	−13.16	−3.09	−0.05	−3.08	−0.05	6.50	−3.05	−0.05
	30	−2.70	−1.05	−0.56	−0.38	0.10	0.99	−0.65	−0.10
	60	−4.74	−1.14	−0.01	−1.28	−0.08	2.31	−1.34	−0.07
0.7	90	−7.82	−1.94	−0.06	−1.98	−0.05	3.66	−2.01	−0.05
	120	−10.81	−2.65	−0.04	−2.65	−0.04	4.97	−2.62	−0.04
	150	−13.78	−3.32	−0.04	−3.26	−0.04	6.25	−3.19	−0.04
MAE		7.78	1.91	0.11	1.99	0.07	3.85	2.06	0.06
ME		−7.78	−1.91	−0.05	−1.99	−0.05	3.85	−2.06	−0.06

This table reports the approximation errors, $e_i=[\log\mathrm{IV}^{\mathrm{model}}_i-\log\mathrm{IV}^{\mathrm{true}}_i]\times 100$ for options with different Moneyness (Black-Scholes Delta), calendar days to maturity (DTM), and initial value of η where $\eta=0.70$ (left panel), $\eta=1.11$ (middle panel), and $\eta=1.70$ (right panel). The true option prices are obtained with 10 million simulated paths from the true model, and the implied volatilities are computed using Black-Scholes formula. The column-wise averages for the mean absolute error (MAE) and the mean error (ME), are presented in the last two rows. CHNG, Auxiliary (AUX), and DHNG correspond to zero-, first-, and second-order approximations, respectively. The model parameters are based on the empirical estimates obtained with option prices (the last column in Table 3).

3. The implied volatility surface depends strongly on moneyness and the level of VIX: out-of-the-money puts are expensive (high implied vol), and implied vols rise sharply in high-VIX regimes.

These features jointly point to time variation in risk-neutral tail probabilities and risk compensation.

Table 2
Summary statistics

A: S&P 500 returns and the CBOE VIX

	Mean(%)	Std(%)	Skewness	Kurtosis	Obs.
Returns (annualized)	8.13	18.11	−0.41	14.37	8,064
VIX	19.48	8.01	2.21	11.49	8,064

B: SPX Option Prices

	Implied Volatility (%)	Average price ($)	Obs.
All options	18.47	63.63	37,152
Partitioned by moneyness			
Delta<0.3	14.18	11.04	5,452
0.3≤Delta<0.4	15.52	20.92	2,955
0.4≤Delta<0.5	16.76	33.92	3,912
0.5≤Delta<0.6	18.94	48.78	7,091
0.6≤Delta<0.7	19.40	64.86	6,098
0.7≤Delta	21.02	117.47	11,644
Partitioned by maturity			
DTM<30	16.89	41.41	9,614
30≤DTM<60	17.91	52.29	9,696
60≤DTM<90	19.00	65.63	7,465
90≤DTM<120	20.25	87.43	4,421
120≤DTM<150	20.22	98.51	2,931
150≤DTM	19.68	97.06	3,025
Partitioned by the level of VIX			
VIX<15	12.15	44.77	12,638
15≤VIX<20	16.70	62.49	10,788
20≤VIX<25	21.45	74.72	7,143
25≤VIX<30	25.43	84.16	3,395
30≤VIX<35	29.61	88.12	1,456
35≤VIX	40.23	101.70	1,732

Summary statistics for close-to-close S&P 500 index log returns, CBOE VIX, and SPX option prices from January 2nd, 1990 to December 31, 2021. We report the sample mean (Mean), standard deviation (Std), skewness (Skew), kurtosis (Kurt), and the number of observations (Obs). Option prices are based on closing prices of out-of-the-money call and put options. We report the average Black-Scholes implied volatility (IV), average price, and the number of option prices for different ranges of Moneyness (Black-Scholes Delta), calendar days to maturity (DTM), and VIX levels. *Sources*: S&P 500 returns from the CRSP and VIX from CBOE's website. Option prices come from Optsum data (1990–1995) and OptionMetrics (1996–2021).

Table 3
Joint estimation results

	CHNG [VIX]	CHNG [Opt]	DHNG [VIX]	DHNG [Opt]
λ	3.377	2.828	3.398	3.088
	(0.661)	(0.022)	(1.140)	(0.644)
β	0.902	0.713	0.720	0.570
	(0.001)	(0.008)	(0.018)	(0.015)
$\alpha(\times 10^{-6})$	1.166	2.495	4.428	5.833
	(0.205)	(0.012)	(0.414)	(0.256)
γ	272.45	327.49	220.45	249.59
	(23.76)	(3.81)	(9.446)	(10.28)
ζ			0.192	0.102
			(0.055)	(0.011)
φ			0.988	0.994
			(0.002)	(0.001)
σ			0.061	0.042
			(0.002)	(0.010)
ρ		0.822		0.087
		(0.015)		(0.010)
σ_e	0.175	0.229	0.043	0.091
	(0.002)	(0.002)	(0.001)	(0.006)
$\eta, \mathbb{E}\eta$	1.371	1.252	1.314	1.190
	(0.071)	(0.013)		
$\widehat{\mathrm{var}}(s_t)$			1.000	1.007
$\pi^{\mathbb{P}}$	0.989	0.981	0.935	0.934
$\pi^{\mathbb{Q}}$	0.991	0.985	0.944	0.944
LogL				
ℓ(R)	26,324	26,438	**26,450**	26,419
ℓ(VIX)	2,619	*1,602*	**13,917**	*7,929*
ℓ(Opt)	*15,177*	21,647	*27,857*	**36,682**
ℓ(R, VIX)	28,944	*28,041*	**40,367**	*34,348*
ℓ(R, Opt)	*41,501*	48,085	*54,307*	**63,101**

Estimation results for the full sample period, January 1990 to December 2021. CHNG is the model by Christoffersen, Heston, and Jacobs (2013) and DHNG is the model introduced in this paper. The type of derivatives used to estimate each model, VIX or option prices, is indicated with [VIX] and [Opt], respectively. Estimates are reported with robust standard errors (in parentheses), and $\pi^{\mathbb{P}}$ and $\pi^{\mathbb{Q}}$ refer to the persistence of volatility under $\mathbb{P}$ and $\mathbb{Q}$, respectively. The components of the maximized log-likelihood function are reported at the bottom of the table, and bold is used to emphasize the largest log-likelihood in each row. Italics are used to point out pseudo-out-of-sample log-likelihoods.

8.5 Table 3: joint estimation and why DHNG dominates CHNG

Table 3 compares CHNG (constant η) vs DHNG (time-varying η_t), estimated using either VIX or option panels. Main lessons:

- **Average variance risk ratio:** η (or $E\eta$) is above one in all specifications, consistent with risk-neutral variance typically exceeding physical variance.
- **Time variation in risk aversion:** DHNG estimates a highly persistent VRR state ($\phi \approx 0.99$) with nontrivial innovation scale σ.

- **Fit improvement:** the derivative-price likelihood terms $\ell(VIX)$ and $\ell(Opt)$ jump dramatically under DHNG, and the total joint likelihood $\ell(R, X)$ is largest for DHNG estimated on options.

- **Smaller measurement errors:** σ_e falls sharply in DHNG, consistent with the state η_t absorbing systematic derivative mispricing that CHNG cannot track.

Interpretation: the data strongly prefer a model that lets volatility risk aversion move over time.

8.6 Figure 5: estimated η_t over 1990–2021

Figure 5 plots daily η_t estimated from VIX vs from option panels. Both series co-move closely:

- η_t spikes in major stress episodes (e.g., the Global Financial Crisis and COVID), indicating sharp increases in volatility risk aversion.

- η_t can dip below one in tranquil periods, consistent with low or even positive variance risk premia.

Interpretation: volatility risk aversion behaves like a state variable that amplifies in crises and relaxes in calm markets.

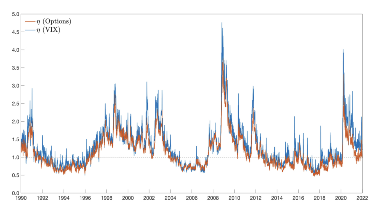

Figure 5
This figure displays the full-sample time series of the daily estimated variance risk ratio, η_t, from January 1990 to December 2021. The blue line is based on the VIX, and the red line on option prices. The two time series are very similar, albeit the η_t based on the VIX tends to be slightly larger than that based on option prices.

Table 4
VIX and option pricing performance

	CHNG [VIX]	CHNG [Opt]	DHNG [VIX]	DHNG [Opt]
A: RMSE for VIX Pricing				
Full sample	17.49	19.84	4.308	9.052
B: RMSE for Option Pricing				
Full sample	23.06	22.88	13.34	9.191
Partitioned by moneyness				
Delta<0.3	33.05	29.80	21.11	12.77
0.3≤Delta<0.4	27.84	26.59	17.86	11.28
0.4≤Delta<0.5	23.29	23.87	14.68	9.375
0.5≤Delta<0.6	19.74	21.49	11.41	7.271
0.6≤Delta<0.7	18.83	20.62	9.72	6.859
0.7≤Delta	19.50	19.48	8.78	8.634
Partitioned by maturity				
DTM<30	25.48	23.51	13.72	12.19
30≤DTM<60	23.48	23.27	12.80	7.848
60≤DTM<90	21.78	22.30	12.75	6.914
90≤DTM<120	20.47	21.81	13.51	7.963
120≤DTM<150	20.74	21.68	13.84	8.622
150≤DTM	22.18	23.58	14.45	9.226
Partitioned by the level of VIX				
VIX<15	29.25	28.06	15.60	10.02
15≤VIX<20	18.47	19.72	13.16	9.021
20≤VIX<25	18.23	19.51	10.89	8.224
25≤VIX<30	18.74	18.52	11.02	8.518
30≤VIX<35	19.48	19.23	10.97	8.542
35≤VIX	25.36	22.41	11.57	9.508

This table presents the full-sample VIX and option pricing performance from January 1990 to December 2021 for each model listed in Table 3. We evaluate the model's VIX and option pricing ability through the root mean square error for log VIX and log implied volatility (logIV), respectively (see Equations (10) and (11)). Results for option pricing are presented for different ranges of moneyness (Black-Scholes Delta), calendar days to maturity (DTM), and the VIX levels.

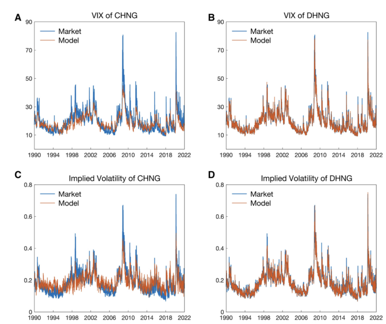

Figure 6
This figure displays the full-sample time series of the daily model-based (red lines) and market-based (blue lines) VIX and average implied volatility. The CHNG and DHNG models are estimated with VIX data in the upper panels and option prices in the lower panels. Model parameters are based on the Table 3.

8.7 Table 4 and Figure 6: in-sample pricing performance

Table 4 reports RMSEs in log VIX (Panel A) and log implied vols (Panel B). The DHNG model reduces pricing RMSEs substantially relative to CHNG, often by a factor of two or more, and especially for VIX. Figure 6 visualizes the same pattern: CHNG systematically over/under-predicts derivative levels depending on volatility regimes, while DHNG tracks the time variation much better.

Economic reading: a constant VRR forces a near-fixed mapping from physical volatility to risk-neutral volatility. The data reject this: the "pricing wedge" between physical and risk-neutral volatility is time-varying, and DHNG captures it via η_t.

8.8 Table 5: out-of-sample pricing performance

Table 5 repeats the exercise out of sample (training on 1990–2007, evaluating 2008–2021). DHNG continues to dominate CHNG for both VIX and option panels. **Interpretation:** the improvement is not just in-sample overfitting; the time-varying risk aversion state generalizes.

8.9 Figure 7: autocorrelation of pricing errors

Figure 7 plots ACFs of pricing errors. CHNG errors exhibit strong persistence (high autocorrelation), consistent with a missing state variable. DHNG largely removes this persistence, indicating that the score-driven η_t absorbs predictable mispricing patterns.

Table 5
Out-of-sample VIX and option pricing

	CHNG [VIX]	CHNG [Opt]	DHNG [VIX]	DHNG [Opt]
A: RMSE for VIX Pricing				
Out-of-sample	18.33	19.60	4.781	10.03
B: RMSE for Option Pricing				
Out-of-sample	24.33	20.98	15.52	9.815
Partitioned by moneyness				
Delta<0.3	36.55	29.30	25.22	13.58
0.3≤Delta<0.4	31.16	25.64	21.19	11.66
0.4≤Delta<0.5	25.81	22.17	17.05	9.540
0.5≤Delta<0.6	20.56	19.05	12.62	7.145
0.6≤Delta<0.7	18.51	17.39	9.373	6.883
0.7≤Delta	16.81	15.81	8.670	9.381
Partitioned by maturity				
DTM<30	28.51	22.73	16.86	11.92
30≤DTM<60	25.27	21.61	15.64	8.372
60≤DTM<90	20.23	18.80	14.07	7.737
90≤DTM<120	19.73	19.22	14.34	8.809
120≤DTM<150	18.02	17.64	14.11	9.539
150≤DTM	21.33	21.02	15.27	10.62
Partitioned by the level of VIX				
VIX<15	32.40	24.86	17.71	10.82
15≤VIX<20	19.31	18.82	15.62	9.580
20≤VIX<25	14.41	18.11	12.99	8.233
25≤VIX<30	16.27	17.42	13.20	9.281
30≤VIX<35	18.40	17.24	13.32	9.569
35≤VIX	27.69	21.02	12.43	9.924

Each model is estimated once, using data from the years 1990–2007 (training sample). The pricing accuracy of each estimated model is evaluated out-of-sample using the remaining 14 years of data, 2008–2021, in terms of the RMSE for log prices. We also present results for subsets of options, partitioning them by moneyness (Black-Scholes Delta), calendar days to maturity (DTM), and the VIX levels.

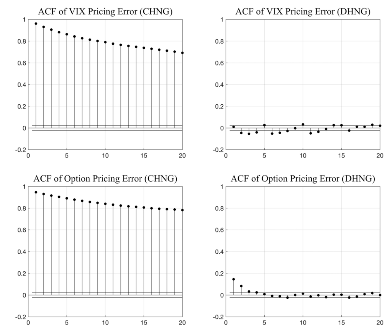

Figure 7
The autocorrelation function (ACF) for derivative pricing errors over the full sample period (January 1990 to December 2021). The ACFs for the VIX are shown in the upper panels, and the ACFs for the average option pricing errors are shown in the lower panels. The CHNG and DHNG models are estimated with VIX data in the upper panels and option prices in the lower panels. Model parameters are based on the Table 3.

Table 6
Economic explanation for time-varying volatility risk aversion
log variance risk ratio (monthly average)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sentiment	–0.095***							–0.093***
	(0.026)							(0.020)
Disp-UNEMP		0.139***						0.008
		(0.052)						(0.044)
Disp-GDP			0.176***					0.050
			(0.040)					(0.049)
EPU				0.380***				0.208***
				(0.060)				(0.045)
SUI					0.148***			0.078***
					(0.026)			(0.025)
EUI						0.071***		0.013
						(0.009)		(0.009)
VRP							0.120***	0.096***
							(0.014)	(0.012)
SVAR	0.395***	0.368***	0.356***	0.294***	0.363***	0.354***	0.395***	0.306***
	(0.041)	(0.043)	(0.043)	(0.039)	(0.042)	(0.041)	(0.029)	(0.029)
DFY	0.358***	0.372***	0.365***	0.393***	0.282***	0.246***	0.285***	0.196***
	(0.091)	(0.088)	(0.087)	(0.078)	(0.087)	(0.081)	(0.067)	(0.058)
DFR	0.027***	0.025***	0.025***	0.019***	0.024***	0.021***	0.026***	0.020***
	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.006)	(0.005)
LTY	0.030**	0.040***	0.043***	0.077***	0.016	0.045***	0.035***	0.063***
	(0.013)	(0.013)	(0.012)	(0.012)	(0.011)	(0.010)	(0.010)	(0.010)
D/P	–0.435***	–0.525***	–0.529***	–0.638***	–0.463***	–0.403***	–0.354***	–0.471***
	(0.127)	(0.130)	(0.129)	(0.117)	(0.133)	(0.122)	(0.091)	(0.085)
TMS	–0.040**	–0.020	–0.018	–0.035**	–0.031*	–0.030*	–0.011	–0.022*
	(0.017)	(0.018)	(0.018)	(0.015)	(0.018)	(0.016)	(0.012)	(0.012)
Other controls	YES	YES	YES	YES	YES	YES	YES	YES
R^2	0.722	0.722	0.732	0.756	0.741	0.761	0.817	0.866
$\rho(1)$	0.579	0.620	0.628	0.612	0.626	0.584	0.326	0.383

The full-sample monthly regression results from January 1990 to December 2021, and the dependent variable is logarithm of monthly averaged variance risk ratio estimated by option prices (the last column of Table 3). The control variables are from Welch and Goyal (2007) and Baker and Wurgler (2006). The column "Other controls" refers to six macro variables that control for macroeconomic conditions and five financial variables. Most of these additional control variables are insignificant (see Internet Appendix IA.4). Newey-West standard errors are reported in parentheses. $\rho(1)$ is the first-order autocorrelation for residuals. Abbreviations: SVAR, stock market volatility; DFY, default yield spread; DFR, default return spread; LTY, long term yield; D/P, dividend price ratio; TMS, term spread. $^{*}p<.1$; $^{**}p<.05$; $^{***}p<.01$.

8.10 Table 6: linking volatility risk aversion to economic fundamentals

Table 6 regresses the monthly average of $\log \eta_t$ (estimated from options) on proxies for disagreement, uncertainty, and sentiment. The robust patterns are:

- **Sentiment:** higher sentiment is associated with *lower* volatility risk aversion (negative coefficient).
- **Uncertainty/disagreement:** measures like dispersion in macro forecasts and policy uncertainty load *positively* on $\log \eta_t$.
- **Variance risk premium:** the VRP proxy is strongly positively related to $\log \eta_t$, consistent with η_t capturing the market price of variance risk.

Interpretation: the estimated volatility risk aversion state is economically meaningful and co-moves with broader measures of uncertainty and sentiment.

9 What to remember

1. The VRR $\eta_t = h^*_{t+1}/h_{t+1}$ is the central state: it translates derivative price information into a time-varying wedge between physical and risk-neutral variance.
2. Time-varying η_t is not just "more flexible": it changes risk-neutral skewness, kurtosis, and tail tilts, and can flip the pricing-kernel shape between U and inverted-U.
3. The paper's analytic approximation (Theorem 3) makes likelihood-based estimation feasible in a non-affine setting.
4. Empirically, allowing η_t to vary dramatically improves pricing of both VIX and option panels, in and out of sample.

5. The estimated η_t loads on uncertainty, disagreement, and sentiment, supporting an interpretation as time-varying volatility risk aversion.

10 Conclusion

10.1 Summary

This paper concludes that a large fraction of the empirical failure of standard discrete-time option-pricing models is not primarily due to the *physical* volatility dynamics, but rather to the restriction that the **compensation for volatility risk is constant**. Allowing **volatility risk aversion** to vary over time—summarized by a single state, the **variance risk ratio**

$$\eta_t \equiv \frac{\mathrm{Var}_t^{\mathbb{Q}}(R_{t+1})}{\mathrm{Var}_t^{\mathbb{P}}(R_{t+1})},$$

produces a model (DHNG) that remains tractable and substantially improves the joint fit to VIX and option panels.

The key conclusions can be stated as one continuous chain:

1. **A flexible pricing kernel can be summarized by a single derivative-implied wedge.** Under the model's assumptions, all time variation in the pricing kernel is captured by η_t (equivalently, a time-varying volatility-risk-aversion parameter). Hence, the mapping from physical volatility to risk-neutral volatility is not fixed, but moves over time.

2. **Time-varying η_t rationalizes the observed instability in pricing-kernel curvature.** The model permits the log pricing-kernel shape (as a function of the return innovation) to change over time, matching empirically observed changes in curvature that a constant-η model cannot reproduce.

3. **Analytical tractability is preserved.** The authors derive (i) a closed-form expression for VIX and (ii) a highly accurate option-pricing approximation that delivers affine-type pricing formulas even though the model becomes non-affine once η_t is stochastic.

4. **Empirically, DHNG sharply reduces derivative pricing errors in and out of sample.** In the full sample, DHNG reduces VIX RMSE dramatically (Table 4, Panel A): e.g.,

$$\begin{aligned}\mathrm{RMSE}_{VIX} &: 17.49 \text{ (CHNG[VIX])} \rightarrow 4.308 \text{ (DHNG[VIX])},\\ &\ 19.84 \text{ (CHNG[Opt])} \rightarrow 9.052 \text{ (DHNG[Opt])}.\end{aligned}$$

   For options (Table 4, Panel B):

$$\mathrm{RMSE}_{Opt}: \ 23.06 \text{ (CHNG[VIX])} \rightarrow 9.191 \text{ (DHNG[Opt])}.$$

   Out of sample (Table 5), the improvements persist and remain economically large:

$$\begin{aligned}\mathrm{RMSE}_{VIX} &: 18.33 \text{ (CHNG[VIX])} \rightarrow 4.781 \text{ (DHNG[VIX])},\\ \mathrm{RMSE}_{Opt} &: 24.33 \text{ (CHNG[VIX])} \rightarrow 9.815 \text{ (DHNG[Opt])}.\end{aligned}$$

   Consistent with the paper's summary, the reduction in derivative pricing RMSE is typically well above 50% once η_t is allowed to vary.

5. **The estimated η_t is economically meaningful.** The time-varying variance risk ratio co-moves with sentiment, disagreement, and uncertainty proxies, and is tightly related to empirical measures of the variance risk premium (Table 6), supporting an interpretation of η_t as a state variable for volatility risk compensation rather than a purely statistical fix.

10.2 Economic intuition: why time-varying volatility risk aversion matters for option prices

The paper's intuition can be stated in three steps.

Step 1: option prices reveal risk-neutral volatility, not physical volatility. Option-implied quantities (VIX and the implied-volatility surface) are functions of risk-neutral expectations of future variance. Thus, matching derivatives requires modeling the *wedge* between $\mathbb{Q}$-variance and $\mathbb{P}$-variance.

Step 2: η_t is the wedge (and it is not constant in the data). A constant-η model forces

$$\mathrm{Var}_t^{\mathbb{Q}}(R_{t+1}) \propto \mathrm{Var}_t^{\mathbb{P}}(R_{t+1}) \quad \text{with a fixed proportionality,}$$

so any time variation in risk-neutral variance must be inherited mechanically from the physical GARCH state. But empirically the option market sometimes demands much more (or much less) risk-neutral variance than the physical dynamics alone can justify. Letting η_t move allows risk-neutral variance to rise sharply in stress periods even if physical variance does not move one-for-one.

Step 3: a score-driven update makes η_t an adaptive "pricing-error absorber" with discipline. The DHNG model updates η_t using the score (a first-order condition) from derivative pricing errors. This has two implications:

- It **eliminates systematic, persistent mispricing** that arises when the true volatility risk compensation changes (visible as strong autocorrelation in CHNG pricing errors).
- It preserves **economic interpretability** because the update is anchored to no-arbitrage restrictions and to an explicit mapping between η_t and the pricing kernel curvature.

This is why DHNG not only reduces RMSE but also reduces the autocorrelation of pricing errors (Figure 7), indicating that the missing state in CHNG is precisely time variation in volatility risk aversion.

10.3 Methodological takeaway

A central methodological conclusion is that **discrete-time GARCH-based option pricing can accommodate a time-varying variance risk ratio without sacrificing tractability**. The paper's approximation method delivers analytic option pricing formulas for a non-affine model and permits likelihood-based estimation using large option panels. The framework is modular: the same time-varying pricing-kernel idea can, in principle, be combined with other GARCH or realized-volatility specifications.

10.4 Takeaway

Option prices are best explained by a time-varying wedge between risk-neutral and physical variance, and modeling this wedge as a score-driven variance risk ratio η_t yields a tractable discrete-time framework that dramatically improves VIX and option pricing and links volatility risk compensation to macro uncertainty and sentiment.